DIPLOMA PROGRAMME PRACTICE · HL · SET B

Chemistry / Paper 1A + 1B

Worked mark scheme · 75 marks · 120 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer A There are 24 protons, so this is a 3+ ion. Adding three electrons neutralizes it without changing the nucleus.

PAPER 1A · QUESTION 2 [1]

Answer D Let fraction 26 be x and fraction 25 be 0.40−x. The mean is 24(0.60)+25(0.40−x)+26x=24.40+x.

PAPER 1A · QUESTION 3 [1]

Answer C The 2p configurations are $2p^3$, $2p^4$, $2p^5$ and $2p^6$, giving 3, 2, 1 and 0 unpaired electrons by Hund’s rule.

PAPER 1A · QUESTION 4 [1]

Answer C Multiply energy per photon by NA to obtain 4.9364×10^5 J mol^{-1}, then divide by 1000.

PAPER 1A · QUESTION 5 [1]

Answer C The 3p orbital in Al is higher in energy than Mg’s 3s. In S the paired electrons repel one another, easing removal compared with P.

PAPER 1A · QUESTION 6 [1]

Answer B At fixed T and p, density is proportional to molar mass. The ratio is (0.400/100)/(0.300/150)=2.00.

PAPER 1A · QUESTION 7 [1]

Answer A Three Mg^{2+} and two N^{3-} balance charges. Displacement can align like charges, whose repulsion splits the lattice.

PAPER 1A · QUESTION 8 [1]

Answer A BF_3 is trigonal planar with three equivalent B–F bonds. The vector sum vanishes, unlike the other molecules.

PAPER 1A · QUESTION 9 [1]

Answer D P forms five bonds with its five valence electrons. I has eight electrons available in the simple count including the extra electron; four bonds leave two lone pairs.

PAPER 1A · QUESTION 10 [1]

Answer A Terminal N: 5−2−3=0; central N: 5−0−4=+1; O: 6−6−1=−1.

PAPER 1A · QUESTION 11 [1]

Answer A Both alkene carbons and the carbonyl carbon have three domains. C=C and C=O each contribute one pi bond.

PAPER 1A · QUESTION 12 [1]

Answer B Each repeat has mass 42, so there are 1000 repeats. Two of the three carbons per repeat lie in the backbone.

PAPER 1A · QUESTION 13 [1]

Answer D The amino acid has both the amino and carboxyl groups needed to form consecutive amide links with loss of water.

PAPER 1A · QUESTION 14 [1]

Answer D Br^- supplies electrons to Cl_2. Each bromide ion loses one electron as elemental bromine forms.

PAPER 1A · QUESTION 15 [1]

Answer C A d^9 subshell contains a vacancy for excitation between split d levels. All available d states are occupied in d^{10}.

PAPER 1A · QUESTION 16 [1]

Answer A Oxidation of secondary butan-2-ol gives butan-2-one, C_4H_8O, by removal of two hydrogen atoms.

PAPER 1A · QUESTION 17 [1]

Answer C An oxygen-bound CH_2 quartet identifies OCH_2CH_3. The singlet at 2.0 ppm is CH_3 adjacent to the carbonyl in $CH_3COOCH_2CH_3$.

PAPER 1A · QUESTION 18 [1]

Answer B The losses are 79 and 81, consistent with the two similarly abundant Br isotopes. Ethyl has nominal mass 29; losing HBr instead would give mass 28.

PAPER 1A · QUESTION 19 [1]

Answer B Each carbon of the double bond must have two different substituents. Both compounds in the correct pair satisfy this condition.

PAPER 1A · QUESTION 20 [1]

Answer D Water gains $150\times4.18\times8.0=5016$ J. Divide 5.016 kJ by 0.400 g.

PAPER 1A · QUESTION 21 [1]

Answer A The gaseous-ion steps sum to +392. Formation requires $+392+\Delta H_{latt}=-350$, hence −742.

PAPER 1A · QUESTION 22 [1]

Answer A Combusting $C+2H_2$ releases $394+572=966$ kJ. Formation of methanol followed by combustion must sum to −966; $\Delta H_f^\circ=-966+726$.

PAPER 1A · QUESTION 23 [1]

Answer C P gives 2.5/50=0.050 and Q gives 1.8/30=0.060 kg CO_2 per MJ of chemical energy.

PAPER 1A · QUESTION 24 [1]

Answer A $\Delta S°=245-360=-115$ J K^{-1} per reaction. $\Delta G°=-32.0-300(-0.115)=+2.50$ kJ.

PAPER 1A · QUESTION 25 [1]

Answer B $\Delta G°=-RT\ln K=-8.31\times310\times\ln(25.0)$ J $mol^{-1}\approx-8.29$ kJ mol^{-1}.

PAPER 1A · QUESTION 26 [1]

Answer B $HCO_3^- + H^+ \rightarrow CO_2 + H_2O$ is 1:1. 0.0100 mol acid at 0.250 mol dm^{-3} occupies 0.0400 dm^3.

PAPER 1A · QUESTION 27 [1]

Answer B Q is first order. The unchanged rate in the combined trial implies P is also first order. Multiply 3 by 0.5.

PAPER 1A · QUESTION 28 [1]

Answer C Step 1 rises 55−0=55. Step 2 starts at −15 and rises to 50, so its barrier is 65.

PAPER 1A · QUESTION 29 [1]

Answer B $E_a=R\ln(2)/(1/300-1/315)=36{,}300$ J mol^{-1}.

PAPER 1A · QUESTION 30 [1]

Answer D Expansion favours two gas particles over one. Heating favours the endothermic reverse reaction.

PAPER 1A · QUESTION 31 [1]

Answer D Remaining HI=0.600 mol. [HI]=0.300, $[H_2]=[I_2]=0.100$, so $K_c=0.0100/0.0900$.

PAPER 1A · QUESTION 32 [1]

Answer B All concentrations double. $Q_c=[C]/([A][B]^2)$ changes by $2/(2\times2^2)=1/4$.

PAPER 1A · QUESTION 33 [1]

Answer D Excess $H^+=0.00100$ mol in 0.0300 dm^3. $pH=-\log_{10}(0.0333)=1.48$.

PAPER 1A · QUESTION 34 [1]

Answer C At half-equivalence the conjugate pair has equal amounts. pH=pKa, so $K_a=10^{-4.65}=2.24\times10^{-5}$.

PAPER 1A · QUESTION 35 [1]

Answer C pOH=2.70. The anion concentration is $10^{-2.70}$ and the hydrogen-ion concentration is $10^{-11.30}$.

PAPER 1A · QUESTION 36 [1]

Answer D Electrode potential is intensive: 0.25−(−0.50)=0.75 V even on doubling. Each of two Y atoms loses two electrons.

PAPER 1A · QUESTION 37 [1]

Answer A Water is reduced to hydrogen and oxidized to oxygen. Sodium and sulfate ions are spectators under the stated conditions.

PAPER 1A · QUESTION 38 [1]

Answer C Adding the two propagation equations cancels both radicals and regenerates Br• as a chain carrier.

PAPER 1A · QUESTION 39 [1]

Answer D For SN1, unimolecular C–Br ionization is rate-determining. The nucleophile reacts after the intermediate forms.

PAPER 1A · QUESTION 40 [1]

Answer B Curly arrows show movement of an electron pair. Alkene pi electrons form the C–H bond while H–Br bonding electrons move to Br.

PAPER 1B · QUESTION 1 [12]

Correcting a combustion measurement

(a) Use the cooling trend to estimate the temperature at shutdown and the corrected temperature rise. [3]

1 Cooling slope is −0.4 K min^{-1}.

1 At 5.0 min the extrapolated temperature is 32.7 °C.

1 Corrected rise is 13.3 K.

(b) Calculate the energy absorbed by water and vessel, and the experimental molar enthalpy of combustion. [3]

1 Combined heat capacity=180.0×4.18+62.0=814.4 J K^{-1}.

1 Heat gained=814.4×13.3=10.83 kJ; n(fuel)=0.620/60.1=0.01032 mol.

1 $\Delta H_{exp} \approx -1.05\times10^{3}$ kJ mol^{-1}.

(c) A reference value for the same fuel and product states is −2020 kJ mol^{-1}. Calculate the percentage of reference energy recovered and explain one chemical and one physical source of deficit. [3]

1 Recovery=1050/2020×100≈52.0%.

1 Incomplete combustion to CO or soot releases less energy per mole of fuel.

1 Heat escapes to the air or unaccounted apparatus while the burner runs, which extrapolation after shutdown cannot fully restore.

(d) Suppose fuel evaporates unburnt during weighing. Predict its effect and propose a paired control to estimate its magnitude. [3]

1 The measured mass loss includes unburnt fuel, so moles burned are overestimated.

1 The apparent combustion enthalpy then has too small a magnitude.

1 Weigh an identical unlit burner over the same handling interval and conditions to estimate evaporation; accept a suitable capped-burner comparison with limitations.

PAPER 1B · QUESTION 2 [12]

Pressure tracking in a sealed cell

(a) Determine the initial amount of N_2O_4 and the fraction that dissociates. [3]

1 $n_0=pV/(RT)=100.0/(8.31\times300.0)=0.04011$ mol.

1 For dissociated fraction α, total moles are $n_0(1+\alpha)$, and pressure is proportional to total moles.

1 $\alpha=130.0/100.0-1=0.300$.

(b) Calculate both equilibrium concentrations and Kc using concentrations in mol dm⁻³. [3]

1 $[N_2O_4]=0.04011\times0.700/1.000=0.02808$ mol dm⁻³.

1 $[NO_2]=2\times0.04011\times0.300/1.000=0.02407$ mol dm⁻³.

1 $K_c=[NO_2]^2/[N_2O_4]=0.0206$ using the stated concentration convention.

(c) Estimate the mean formation rate of NO_2 from 20 to 80 s. Explain the factor linking increase in total moles to moles of NO_2 formed. [3]

1 $\Delta n_{total}=V\Delta p/(RT)=15/(8.31\times300)=0.00602$ mol.

1 One mole dissociated increases total moles by one but forms two moles NO_2; $\Delta n(NO_2)=0.0120$ mol.

1 Mean rate $=0.0120/(1.000\times60)=2.01\times10^{-4}$ mol dm⁻³ s⁻¹.

(d) A student interprets a rising pressure as sufficient proof of dissociation. Give a competing explanation and an experimental check. Explain why a plateau alone does not prove dynamic equilibrium. [3]

1 Heating a sealed fixed amount of gas can also raise its pressure.

1 Monitor the gas temperature independently in a thermostatted bath and test the vessel for leaks.

1 A flat reading shows constant pressure within resolution; ongoing equal forward and reverse rates require chemical evidence or consistency with reversible changes.

PAPER 1B · QUESTION 3 [11]

A cell under load

(a) Write the net ionic reaction and calculate the standard potential and Gibbs energy change per mole of Zn. [3]

1 $Zn(s)+Cu^{2+}(aq)\rightarrow Zn^{2+}(aq)+Cu(s)$.

1 $E^\circ cell=0.34-(-0.76)=1.10$ V.

1 $\Delta G^\circ=-2\times96{,}500\times1.10=-212$ kJ mol^{-1}.

(b) Compare the two open-circuit readings with each other using maximum absolute uncertainties. Is the decline resolved? [3]

1 The measured fall is 0.01 V.

1 Maximum uncertainty in the difference is $0.01+0.01=0.02$ V.

1 Zero difference is possible within this range, so a decline is not resolved.

(c) Explain why the loaded voltage is unsuitable for a direct calculation of standard Gibbs energy and identify the direction of electron flow through the resistor. [2]

1 A current produces internal voltage losses/polarization, so the terminal voltage under load is not the reversible standard emf.

1 Electrons flow from Zn, the oxidation electrode, toward Cu.

(d) A technician replaces the salt bridge with a dry insulating strip. Explain the resulting behaviour and propose a control that tests the bridge rather than the electrodes. [3]

1 Charge imbalance builds up in the half-cells without ion migration.

1 Sustained external electron flow ceases even if a brief transient is seen.

1 Restore the original conducting salt bridge with electrodes and solutions unchanged and check whether the normal current/voltage behaviour returns.